Project: Worked examples
Title: ISO 286 limits and fits
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Hole and shaft limits for a nominal size and tolerance classes, the resulting fit (clearance, transition or interference) and its extreme values, from the ISO 286-1 formulae.
i = 0.45·D^(1/3) + 0.001·D (µm) at the geometric mean D of the size step; ITn = multiplier × i; fundamental deviations by letter from the published formulae; hole deviations by the general rule (EI = −es for A–H; ES = −ei + Δ for K–ZC). Limits = nominal + deviations; fit extremes from the limits. A rolling-bearing bore is not an ISO 286 hole: it has its own tolerance (ISO 492) lying below nominal, so in that mode ES and EI are entered from the bearing record.
Ref: ISO 286-1:2010 §4, §5 — Bases of tolerances, deviations and fits — tolerance grades and fundamental deviations
Ref: ISO 286-2:2010 — Tables of standard tolerance classes and limit deviations (not reproduced)

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
D	Nominal size	50	mm	drawing: DRG-HUB-04 rev C	1 … 500 mm
hole_kind	Hole: an ISO 286 class, or a rolling-bearing bore (ES and EI from the bearing record, ISO 492)	ISO 286 hole class		default	any
hole_letter	Hole fundamental deviation letter	H		spec: DRG-HUB-04: H7/p6	any
hole_grade	Hole IT grade	7		spec: DRG-HUB-04: H7/p6	5 … 11
shaft_letter	Shaft fundamental deviation letter	p		spec: DRG-HUB-04: H7/p6	any
shaft_grade	Shaft IT grade	6		spec: DRG-HUB-04: H7/p6	4 … 11

3. Assumptions

A1. IT values and fundamental deviations are the unrounded results of the ISO 286-1 formulae at the geometric mean of the size step. (The tabulated values apply the standard's own rounding and may differ by about 1 µm; enter overrides from a released reference record where the tabulated value governs.)
A2. Hole deviations follow the general rule from the shaft deviations, with Δ = ITn − IT(n−1) where the standard applies it. (Table-specific departures in the standard are not reproduced; use the overrides where they govern.)

4. Working

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[7] D_step = size step (lower, upper]
= (30, 50] mm
= 38.73 mm
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ISO 286-1 §4

Geometric mean of the size step

[8] i = 0.45·D^(1/3) + 0.001·D ISO 286-1 §5
= D = 38.73 mm
= 1.5612 µm
Standard tolerance unit
[9] IT_hole = IT7 ISO 286-1 §5
= 16 × i
= 24.98 µm
Hole tolerance
[10] IT_shaft = IT6 ISO 286-1 §5
= 10 × i
= 15.612 µm
Shaft tolerance
[11] es = ei + IT_shaft ISO 286-1 §5
= 24.98 + 15.612
= 40.592 µm
Shaft upper deviation
[12] ei = fundamental deviation 'p' ISO 286-1 §5
= +IT7
= 24.98 µm
Shaft lower deviation
[13] ES = EI + IT_hole ISO 286-1 §5
= 0 + 24.98
= 24.98 µm
Hole upper deviation
[14] EI = fundamental deviation 'H' ISO 286-1 §5
= -es of the same letter
= 0 µm
Hole lower deviation
[15] D_hole_max = D + ES
= 50 mm + 24.98 µm
= 50.025 mm
Hole maximum
[16] D_hole_min = D + EI
= 50 mm + 0 µm
= 50 mm
Hole minimum
[17] d_shaft_max = D + es
= 50 mm + 40.592 µm
= 50.041 mm
Shaft maximum
[18] d_shaft_min = D + ei
= 50 mm + 24.98 µm
= 50.025 mm
Shaft minimum
[19] c_max = ES - ei
= 24.98 µm − 24.98 µm
= 0 µm
Maximum clearance (negative: minimum interference)
[20] c_min = EI - es
= 0 µm − 40.592 µm
= -40.592 µm
Minimum clearance (negative: maximum interference)

[21] i_max = es - EI
= 40.592 µm − 0 µm
= 40.592 µm
Maximum interference
[22] i_min = ei - ES
= 24.98 µm − 24.98 µm
= 0 µm
Minimum interference (negative: maximum clearance)
[23] fit = fit class
= H7/p6
= "interference"
Fit kind: clearance, transition or interference
[24] c_mean = (c_max + c_min) / 2
= (0 µm + -40.592 µm) / 2
= -20.296 µm
Mean clearance

5. Results

SYMBOL	VALUE	DESCRIPTION
IT_hole	24.98 µm	Hole tolerance
IT_shaft	15.612 µm	Shaft tolerance
ES	24.98 µm	Hole upper deviation
EI	0 µm	Hole lower deviation
es	40.592 µm	Shaft upper deviation
ei	24.98 µm	Shaft lower deviation
D_hole_max	50.025 mm	Hole maximum
D_hole_min	50 mm	Hole minimum
d_shaft_max	50.041 mm	Shaft maximum
d_shaft_min	50.025 mm	Shaft minimum
fit	"interference"	Fit kind
c_max	0 µm	Maximum clearance
c_min	-40.592 µm	Minimum clearance
i_max	40.592 µm	Maximum interference
i_min	0 µm	Minimum interference
c_mean	-20.296 µm	Mean clearance

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Fit classified	H7/p6	interference	0	

7. Applicability

WARNING — input.hole_kind: hole_kind (Hole: an ISO 286 class, or a rolling-bearing bore (ES and EI from the bearing record, ISO 492)) was not entered; the template default 'ISO 286 hole class' is used.

8. Verification

Template mech.iso286-fit@1.1.0 Content hash 5a671a70cd810125 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0